

RAG + Generative AI Service – API Reference

Version 1.0 | Platform Engineering

This document defines the HTTP API exposed by the RAG + Generative AI

Service. All endpoints are served over HTTPS. JSON is the only supported request and response format.

Base URL: `https://api.company.example`

Authentication:

- Bearer token in the Authorization header.
- Machine accounts use OAuth 2.0 client credentials.
- Requests without a valid token return HTTP 401.

Headers:

- X-Request-Id: client-generated correlation id (echoed in responses).
- Content-Type: application/json on all POST bodies.

All timestamps use ISO 8601 with a UTC offset.

Endpoints

GET /healthz

Returns service status, model, retrieval mode and index size.

Response 200: {"status": "ok", "model": "...", "retrieval_mode": "hybrid", "index_size": 218}

POST /v1/chat

Multi-turn chat completion grounded in the knowledge base.

Request body: {"session_id": "s1", "messages": [{"role": "user", "content": "□□□□□□□□"}]}

Response 200: answer, citations, refusal, usage and latency.

A refusal includes a refusal_reason field and guidance text.

POST /v1/debug

Full pipeline trace for diagnostics. Returns the retrieved candidates, reranked top-k, the exact prompt sent to the model and the answer.

GET /v1/chunks

Lists every chunk in the vector index. Optional ?source=filter.

Response: {"total": 218, "sources": [...], "chunks": [...]}

Request and Response Examples

Example 1 – chat request:

POST /v1/chat

```
{"session_id": "s1",  
"messages": [{"role": "user",  
"content": "□□□□□□□□"}]}
```

Response:

```
{"request_id": "req_abc123",  
"answer": "□□□□□□□□ 15 □□□□□□",  
"citations": [{"chunk_id": "employee-handbook.cn.md#4"}],  
"refusal": null, "grounded": true,  
"usage": {"input_tokens": 2382, "output_tokens": 28},  
"latency_ms": 1420.5}
```

Example 2 – refusal:

```
{"refusal": "refused",  
"refusal_reason": "out_of_scope",  
"answer": "□□□□□□□□□□□□□□□□"}
```

Personal data in answers is redacted with placeholders such as <PII:EMAIL>.

Error Codes, Rate Limits and Pagination

Error envelope:

```
{"error": {"code": "validation_error",  
"message": "messages must not be empty"}}
```

Common codes:

401 authentication_failed

403 forbidden

404 not_found

409 conflict

422 validation_error

429 rate_limited

500 internal_error

Rate limiting:

- Public endpoints are limited to 120 requests per minute per client.
- Exceeded limits return 429 with a Retry-After header.

Pagination:

- List endpoints use cursor-based pagination.
- Page size is capped at 100 by default.

Grounding policy:

- Answers must cite the retrieved chunk ids.
- When context is insufficient the service refuses instead of guessing.
- Model outputs are grounded and PII is redacted.